Lección 18

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May 24, 2026

Plan de trabajo

- **Formulación hamiltoniana**
  - Ecuaciones de Hamilton
  - Transformaciones canónicas
- **Nuestro Syllabus:**

FIS-310:

UNIDAD 5. Formulación Hamiltoniana

5.1. Transformaciones de Legendre

5.2. Ecuaciones de Hamilton

5.3. Ecuaciones de Hamilton de la Partícula Relativista

5.4. Corchete de Poisson

5.5. Transformaciones Canónicas

5.6. Ecuaciones de Hamilton-Jacobi

5.7. Teoría de perturbaciones

FIS-345:

Unidad IV: Formulación Hamiltoniana General

- Transformaciones canónicas
- Variables de ángulo y acción. Invariantes Adiabáticos

Unidad V: Teoría de perturbaciones y pequeñas oscilaciones

- Teoría canónica de perturbaciones
- Caos en sistemas Hamiltonianos

FIS-426:

4) Formulación Hamiltoniana General

- Paréntesis de Poisson
- Transformaciones canónicas
- Variables de ángulo y acción
- Teoría de perturbaciones
- Caos en sistemas Hamiltonianos

Before we start ...

Please try to recall (from undergraduate courses) what is canonical transformation:

– Formal definition ...

– What conditions it should satisfy? How to construct it?

– Why do we need these "canonical transformation"?

Hamiltonian formulation (summary)

- Use momenta $\boldsymbol{p}_a$ and coordinates $\boldsymbol{q}_a$ to describe system

—"vector" in phase space

- Dynamics is described by Hamiltonian H, related to L via Legendre's transformation

$$H \equiv \sum p_a \dot{q}_a - L, \qquad L = \sum_a p_a \frac{\partial H}{\partial p_a} - H, \qquad p = \frac{\partial L}{\partial \dot{q}_a}$$

Hamilton's (Canonical) equations

$$\dot{p}_a = -\frac{\partial H(p, q, t)}{\partial q_a} = [H, p_a], \qquad \dot{q}_a = \frac{\partial H(p, q, t)}{\partial p_a} = [H, q_a], \qquad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.$$

If the particle follows the trajectory $\{p_a(t), q_a(t)\}$, then change of *any* function $f(p, q, t)$ along the trajectory is

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_a \left(\frac{\partial f}{\partial q_a}\dot{q}_a + \frac{\partial f}{\partial p_a}\dot{p}_a \right) = \frac{\partial f}{\partial t} + \sum_a \left(\frac{\partial H}{\partial p_a}\frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a}\frac{\partial f}{\partial p_a} \right) = \frac{\partial f}{\partial t} + [H, f]$$

$$[H, f] \equiv \sum_a \left(\frac{\partial H}{\partial p_a}\frac{\partial f}{\partial q_a} - \frac{\partial H}{\partial q_a}\frac{\partial f}{\partial p_a} \right) \qquad (= \text{Poisson bracket})$$

$\Rightarrow H$ plays the role of generator for translations $t \to t + t_0$ (recall ***Lecture 13***)

Canonical equations - symplectic notations

$$\boldsymbol{\eta} = \begin{pmatrix} q_1 \\ \dots \\ q_N \\ p_1 \\ \dots \\ p_N \end{pmatrix}, \qquad \Rightarrow \quad \mathrm{EOM}: \quad \dot{\boldsymbol{\eta}} = [H, \boldsymbol{\eta}] = \boldsymbol{J} \frac{\partial H}{\partial \boldsymbol{\eta}}, \quad [\eta_a, \eta_b] = J_{ab}$$

$$J = \begin{pmatrix} & I_N \\ -I_N & \end{pmatrix} = \begin{pmatrix} & & & 1 & 0 & 0 \\ & 0 & & 0 & \dots & 0 \\ & & & 0 & 0 & 1 \\ -1 & 0 & 0 & & & \\ 0 & \dots & 0 & & 0 & \\ 0 & 0 & -1 & & & \end{pmatrix}$$

– The peculiar antisymmetric structure of J_{ab} should be the same after all possible transformations of coordinates, in all inertial reference frames
  * **Canonical transformations**: all transformations of η which **don't** change the structure of J_{ab}

Canonical transformations (motivation)

Earlier discussion:

▶ Sometimes non-Cartesian coordinates simplify drastically analysis of the problem

– Example: Kepler problem→spherical coordinates

▶ Important fact which justifies use of curvilinear coords:

– Covariance of Euler-Lagrange equations with respect to substitutions of variables $q_a \to q_a(r_1, r_2, r_3)$

$$\frac{\partial L}{\partial q_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right) = 0 \quad \Leftrightarrow \quad \frac{\partial L}{\partial r_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}_a}\right) = 0$$

▶ **Hamiltonian approach**:

– Coordinates and momenta contribute on equal footing, as "coordinates" $\boldsymbol{\eta} = \{q_a, p_a\}$ in phase space

* Transformations Cartesian↔spherical↔... mix only coordinates q_a (first N components of $\boldsymbol{\eta}$) with each other

* Potentially could also consider transformations which also mix coordinates q_a and momenta p_a ???

Canonical transformations (introduction)

Caution:

► We still want to have the same form of canonical equations

$$\dot{P}_a = -\frac{\partial \tilde{H}(P, Q, t)}{\partial Q_a}, \qquad \dot{Q}_a = \frac{\partial \tilde{H}(P, Q, t)}{\partial P_a}, \qquad (1)$$

or in symplectic form

$$\dot{\boldsymbol{\eta}} = [H, \boldsymbol{\eta}] = \boldsymbol{J}\frac{\partial H}{\partial \boldsymbol{\eta}}, \qquad \boldsymbol{J} = \begin{pmatrix} & I_N \\ -I_N & \end{pmatrix} \qquad (2)$$

in order to be able to interpret new variables P, Q as coordinates and momenta

► If we consider *arbitrary* transformation

$$\begin{aligned} Q_a &= Q_a(q_1, ..., q_N, p_1, ..., p_N, t), \\ P_a &= P_a(q_1, ..., q_N, p_1, ..., p_N, t). \end{aligned}$$

then clearly (1,2) might be not valid (get extra factors with derivatives).

Canonical transformations

Canonical transformation

Transformation $(q,\ p,\ t) \to (Q,\ P,\ t)$ is canonical if it preserves the form of canonical (Hamilton) equations, $i.e.$

$$\dot{P}_a = -\frac{\partial \tilde{H}(P,\ Q, t)}{\partial Q_a}, \qquad \dot{Q}_a = \frac{\partial \tilde{H}(P,\ Q, t)}{\partial P_a}, \qquad (1)$$

►**Some trivial properties**:

−The identity transformation $\{P_a = p_a,\ Q_a = q_a\}$ is canonical

−If the transformation is canonical, then its inverse is also canonical

−Superposition of two canonical transformations is also canonical

Lemma

Earlier we discussed special (limited) case of transformations which affects only coordinates,

$$q_\alpha \to q_\alpha\left(Q_1, ..., Q_N\right), \qquad Q_\alpha \to Q_\alpha\left(q_1, ..., q_N\right)$$

don't mix them with momenta, e.g. Cartesian$\leftrightarrow$Spherical$\leftrightarrow$Cylindrical. We'll call them ***point*** transformations

Lemma

All *point* transformations of coordinates are canonical

In simple words:
We know that lagrangian formalism and Euler-Lagrange equations are valid in any curvilinear coordinates (spherical, polar, Cartesian). For this reason we expect that Hamiltonian approach should give equivalent results in any of these frames. Now we need to show that this is indeed so.

Proof of lemma

– Assume (q_a, p_a) are variables before transformation, and (Q_a, P_a) are variables after tranformation

– Canonical equations:

$$\dot{q}_a = \frac{\partial H}{\partial p_a},\ \dot{p}_a = -\frac{\partial H}{\partial q_a}$$

– Apply chain rule to the first equation, recalling that $q_\alpha\left(Q_1, ..., Q_N\right)$

$$\dot{q}_a = \sum_b \dot{Q}_b \frac{\partial q_a}{\partial Q_b} = \frac{\partial H}{\partial p_a} \qquad (1)$$

– Now recall that $Q_\alpha\left(q_1, ..., q_N\right)$ and work with chain rule again:

$$\delta_b^c = \frac{\partial Q_c}{\partial Q_b} = \sum_b \frac{\partial q_a}{\partial Q_b} \frac{\partial Q_c}{\partial q_a} = \delta_b^c \qquad (2)$$

$$\sum_{ab} \dot{Q}_b \frac{\partial q_a}{\partial Q_b} \frac{\partial Q_c}{\partial q_a} = \sum_b \dot{Q}_b \delta_b^c = \dot{Q}_c \qquad (3)$$

Now let's multiply both parts of (1) on matrix $\frac{\partial Q_c}{\partial q_a}$ and make summation over a (the idea is to try to extract $\dot{Q}_a$ using (2,3))

Proof of lemma (II)

$$\Rightarrow \dot{Q}_c = \sum_a \frac{\partial H}{\partial p_a} \left(\frac{\partial Q_c}{\partial q_a} \right) := \frac{\partial H}{\partial P_c} \qquad (1)$$

–Expression in right-hand side is what we *expect* for point transformation. If we *assume* that momenta P_a are related to momenta p_a as

$$\frac{\partial P_a}{\partial p_b} = \left(\frac{\partial q_b}{\partial Q_a} \right), \quad \frac{\partial p_a}{\partial P_b} = \frac{\partial Q_b}{\partial q_a} \qquad (2)$$

then (1) turns into chain rule, becomes valid identically

Proof of lemma (III)

$$\frac{\partial P_a}{\partial p_b} = \left(\frac{\partial q_b}{\partial Q_a}\right), \quad \frac{\partial p_a}{\partial P_b} = \frac{\partial Q_b}{\partial q_a} \qquad (2)$$

−May repeat the same exercise for the second canonical equation:

$$\dot{p}_a = -\frac{\partial H}{\partial q_a} = \sum_b -\frac{\partial H}{\partial Q_b}\frac{\partial Q_b}{\partial q_a} \qquad (3)$$

Let's rewrite (2) as

$$\dot{P}_a = \left(\frac{\partial q_b}{\partial Q_a}\right)\dot{p}_b, \quad \Rightarrow \dot{p}_b = \left(\frac{\partial Q_b}{\partial q_a}\right)\dot{P}_b \qquad (2')$$

substitute (2') into (3) and multiply both parts onto matrix $\partial q_b/\partial Q_a$:

$$\Rightarrow \dot{P}_b = -\frac{\partial H}{\partial Q_b}$$

as expected for canonical transformation

⇒If transformation of momenta $P(p)$ agrees with (2), then we can guarantee that point transformation is canonical, Q.E.D.

Canonical transformations of coordinates

Assume that we have a general transformation

$$q = q\,(Q,\, P)\,, \quad p = p\,(Q,\, P)\,.$$

What conditions it should satisfy in order to be canonical?

Please suggest algorithm which could be used to find such relations.

Canonical transformations of coordinates

$$q = q\left(Q,\, P\right), \quad p = p\left(Q,\, P\right).$$

By definition, canonical transformation should NOT change the form of canonical equations

$$\Downarrow$$

Our **algorithm of derivation**:

1) Write the canonical equations in coordinates (q, p)

$$\dot{p}_a = -\frac{\partial H}{\partial q_a}, \qquad \dot{q}_a = \frac{\partial H}{\partial p_a},$$

2) Rewrite derivatives in terms of variable (Q, P) using chain rules for derivatives
3) Request that all "extra" factors cancel, so that get normal canonical equations in terms of variables (Q, P)

$$\dot{P}_a = -\frac{\partial H\left(P,\, Q, t\right)}{\partial Q_a}, \qquad \dot{Q}_a = \frac{\partial H\left(P,\, Q, t\right)}{\partial P_a},$$

–Subtle point: we assume that hamiltonian does not change, *i.e.* just need to substitute variables in "old" hamiltonian (*this is correct for time-independent transformations*):

$$H_{PQ}(P, Q) = H_{pq}\left(q = q\left(Q,\, P\right),\, p = p\left(Q,\, P\right)\right)$$

Canonical transformations of coordinates

$$\dot{p}_a = -\frac{\partial H}{\partial q_a}, \quad \dot{q}_a = \frac{\partial H}{\partial p_a}, \qquad \dot{P}_a = -\frac{\partial H}{\partial Q_a}, \quad \dot{Q}_a = \frac{\partial H}{\partial P_a}$$

$$\frac{\partial H}{\partial P_a} = \frac{\partial H}{\partial p_b}\frac{\partial p_b}{\partial P_a} + \frac{\partial H}{\partial q_b}\frac{\partial q_b}{\partial P_a}$$

$$\dot{Q}_a = \frac{\partial H}{\partial P_a} = \underbrace{\frac{\partial H}{\partial p_b}}_{\dot{q}_b}\frac{\partial p_b}{\partial P_a} + \underbrace{\frac{\partial H}{\partial q_b}}_{-\dot{p}_b}\frac{\partial q_b}{\partial P_a}$$

$$\dot{Q}_a = \frac{\partial Q_a}{\partial p_b}\,\dot{p}_b + \frac{\partial Q_a}{\partial q_b}\,\dot{q}_b$$

$$\Rightarrow \left(\frac{\partial q_b}{\partial P_a}\right)_{Q,P} = -\left(\frac{\partial Q_a}{\partial p_b}\right)_{q,p}, \qquad \left(\frac{\partial Q_a}{\partial q_b}\right)_{p,q} = \left(\frac{\partial p_b}{\partial P_a}\right)_{Q,P}$$

Similar analysis of the equation for $\dot{P}_a$ yields

$$\left(\frac{\partial P_b}{\partial p_a}\right)_{q,p} = \left(\frac{\partial q_a}{\partial Q_b}\right)_{Q,P}, \qquad \left(\frac{\partial P_a}{\partial q_b}\right)_{p,q} = -\left(\frac{\partial p_b}{\partial Q_a}\right)_{Q,P}$$

Lemma

Poisson bracket $[f, g]$

$$[f, g]_{p,q} = \sum_a \left(\frac{\partial f}{\partial p_a} \frac{\partial g}{\partial q_a} - \frac{\partial f}{\partial q_a} \frac{\partial g}{\partial p_a} \right)$$

$$[f, g]_{P,Q} = \sum_a \left(\frac{\partial f}{\partial P_a} \frac{\partial g}{\partial Q_a} - \frac{\partial f}{\partial Q_a} \frac{\partial g}{\partial P_a} \right)$$

If $(p, q) \to (P, Q)$ is canonical transformation, then Poisson bracket is invariant, i.e.

$$[f, g]_{p,q} = [f, g]_{P,\,Q}$$

Canonical transformations of coordinates

Demonstrate that if $(p, q) \to (P, Q)$ is canonical transformation, then Poisson bracket is invariant, i.e.

$$[f, g]_{p,q} = [f, g]_{P, Q}$$

$$[f, g]_{p,q} = \sum_a \left(\frac{\partial f}{\partial p_a} \frac{\partial g}{\partial q_a} - \frac{\partial f}{\partial q_a} \frac{\partial g}{\partial P_a} \right)$$

$$\frac{\partial f}{\partial p_a} = \sum_b \left(\frac{\partial f}{\partial Q_b} \frac{\partial Q_b}{\partial p_a} + \frac{\partial f}{\partial P_b} \frac{\partial P_b}{\partial p_a} \right)$$

$$\frac{\partial f}{\partial q_a} = \sum_b \left(\frac{\partial f}{\partial Q_b} \frac{\partial Q_b}{\partial q_a} + \frac{\partial f}{\partial P_b} \frac{\partial P_b}{\partial q_a} \right)$$

$$[f, g]_{p,q} =$$

$$\sum_{abc} \left(\frac{\partial f}{\partial P_b} \frac{\partial g}{\partial Q_c} - \frac{\partial g}{\partial P_b} \frac{\partial f}{\partial Q_c} \right) \frac{\partial P_b}{\partial p_a} \frac{\partial Q_c}{\partial q_a}$$

but we have seen that

$$\left(\frac{\partial Q_a}{\partial q_b} \right)_{p,q} = \left(\frac{\partial p_b}{\partial P_a} \right)_{Q,P}$$

$$\left(\frac{\partial P_a}{\partial q_b} \right)_{p,q} = - \left(\frac{\partial p_b}{\partial Q_a} \right)_{Q,P}$$

$$\Rightarrow \frac{\partial P_b}{\partial p_a} \frac{\partial Q_c}{\partial q_a} = \frac{\partial P_b}{\partial p_a} \frac{\partial p_c}{\partial P_c} = \delta_{bc}$$

$$\Rightarrow [f, g]_{p,q} = [f, g]_{P,Q}$$

► We conclude that Poisson bracket does not change after canonical transformation

Some consequences

$$[f,\, g]_{p,q} = [f,\, g]_{P,\, Q}$$

Why this result is important?

►We know that

$$[q_a,\, q_b]_{p,q} = 0, \quad [p_a,\, p_b]_{p,q} = 0, \quad [p_a,\, q_b]_{p,q} = \delta_{ab},$$

and

$$[Q_a,\, Q_b]_{P,Q} = 0, \quad [P_a,\, P_b]_{P,Q} = 0, \quad [P_a,\, Q_b]_{P,Q} = \delta_{ab},$$

►Assume we know $P_a = P_a(p,\, q)$ and $Q_b = Q_b(p,\, q)$. Then for canonical transform

$$[Q_a,\, Q_b]_{p,\, q} = 0, \quad [P_a,\, P_b]_{p,q} = 0, \quad [P_a,\, Q_b]_{p,q} = \delta_{ab},$$

-usually very easy to check

Example

Demonstrate that the transformation

$$Q = \log\left(\frac{\sin p}{q}\right), \quad P = q\cot p$$

is canonical

- $[Q,\, Q]_{p,q} = 0$ and $[P,\, P]_{p,q} = 0$ due to antisymmetry of Poisson's bracket
-

$$[P,\, Q]_{p,q} = \frac{\partial P}{\partial p}\frac{\partial Q}{\partial q} - \frac{\partial P}{\partial q}\frac{\partial Q}{\partial p} = -\frac{\cancel{q}}{\sin^2 p}\,\frac{-1}{\cancel{q}} - \cot p\,\cot p =$$
$$= \left(\frac{1}{\sin^2 p} - \cot^2 p\right) = 1.$$

This finishes the proof

Example 2

Using properties of Poisson's brackets, demonstrate that Galileo's transformation is canonical

$$q_a = Q_a + v_a t, \quad p_a = P_a + m\, v_a$$

- Just evaluate brackets assuming $v_a = \mathrm{const}$

$$[q_a,\, q_b]_{P,Q} = [Q_a + v_a t,\, Q_b + v_b t]_{P,Q}$$
$$= [Q_a,\, Q_b]_{P,Q} + t^2\,[v_a,\, v_b]_{P,Q} + t\,[Q_a,\, v_b]_{P,Q} + t\,[v_a,\, Q_b]_{P,Q} = 0$$

$$[p_a,\, p_b]_{P,Q} = [P_a + m\, v_a,\, P_b + m\, v_b]_{P,Q}$$
$$= [P_a,\, P_b]_{P,Q} + m^2\,[v_a,\, v_b]_{P,Q} + m\,[P_a,\, v_b]_{P,Q} + m\,[v_a,\, P_b]_{P,Q} = 0$$

$$[p_a,\, q_b]_{P,Q} = [P_a + m\, v_a,\, Q_b + v_b t]_{P,Q}$$
$$= \underbrace{[P_a,\, Q_b]_{P,Q}}_{\delta_{ab}} + m\, t\,[v_a,\, v_b]_{P,Q} + t\,[P_a,\, v_b]_{P,Q} + m\,[v_a,\, Q_b]_{P,Q} = \delta_{ab}$$

Thus we have shown that Poisson's brackets satisfy the required condition, which proves that transformation is canonical

Example 3

Symplectic form (definition)

The 2-form **defined** as

$$\omega = \sum dq^a \wedge dp^a \equiv -\sum dp^a \wedge dq^a \equiv \sum J_{ab} d\eta^a \otimes d\eta^b = \frac{1}{2} \sum J_{ab} d\eta^a \wedge d\eta^b$$

$$J \equiv \begin{pmatrix} & I_N \\ -I_N & \end{pmatrix}$$

Problem

Demonstrate that for canonical transformation the symplectic formω is invariant, i.e.

$$\omega = \sum dq^a \wedge dp^a = \sum dQ^b \wedge dP^b$$

Example 3 (proof)

Start with definition:

$$\omega = \sum dq^a \wedge dp^a$$

$$dq^a = \frac{\partial q^a}{\partial Q^c} dQ^c + \frac{\partial q^a}{\partial P^c} dP^c = \frac{\partial P^c}{\partial p^a} dQ^c - \frac{\partial Q^c}{\partial p^a} dP^c$$

$$dp^a = \frac{\partial p^a}{\partial Q^e} dQ^e + \frac{\partial p^a}{\partial P^e} dP^e = -\frac{\partial P^e}{\partial q^a} dQ^e + \frac{\partial Q^e}{\partial q^a} dP^e$$

When making substitution, should take into account that

$$dQ^c \wedge dQ^e = -dQ^e \wedge dQ^c, \qquad dP^c \wedge dP^e = -dP^e \wedge dP^c,$$

The constants in front of $dQ \wedge dQ$ and $dP \wedge dP$, given by

$$\frac{\partial P^c}{\partial p^a} \frac{\partial P^e}{\partial q^a}, \qquad \frac{\partial Q^c}{\partial p^a} \frac{\partial Q^e}{\partial q^a}$$

vanish after antisymmetrization w.r.t. $c \leftrightarrow e$:

$$\frac{\partial P^c}{\partial p^a} \frac{\partial P^e}{\partial q^a} - \frac{\partial P^e}{\partial p^a} \frac{\partial P^c}{\partial q^a} = [P^c, P^e] = 0, \qquad \frac{\partial Q^c}{\partial p^a} \frac{\partial Q^e}{\partial q^a} - \frac{\partial Q^e}{\partial p^a} \frac{\partial Q^c}{\partial q^a} = [Q^c, Q^e] = 0$$

so the terms $dQ \wedge dQ$ and $dP \wedge dP$ won't appear. For the constant in front of $dP^c \wedge dQ^e$ we get

$$\frac{\partial q^a}{\partial Q^e} \frac{\partial Q^c}{\partial q^a} + \frac{\partial p^a}{\partial Q^e} \frac{\partial Q^c}{\partial p^a} = \frac{\partial Q^c}{\partial Q^a} = \delta^c_a$$

Symplectic group

The symplectic form

$$\omega = \sum_a dq^a \wedge dp^a = \sum_{ab} J_{ab} d\eta^a \wedge d\eta^b$$

plays crucial role in formulation of the hamiltonian formalism.

$-\omega$ must be invariant under canonical transformation

** Mathematics: All transformations which have this property form a group $\mathrm{Sp}\left(2n, \mathbb{R}\right)$

$$\forall M \in \mathrm{Sp}\left(2n, \mathbb{R}\right), \quad M^T J M = J, \; J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$$

$$\forall \xi, \eta, \quad [M\boldsymbol{\xi}, \; M\boldsymbol{\eta}] = [\boldsymbol{\xi}, \; \boldsymbol{\eta}]$$

Symplectic form

$$\omega = \sum dq^a \wedge dp^a \equiv -\sum dp^a \wedge dq^a \equiv \sum J_{ab} d\eta^a \otimes d\eta^b = \frac{1}{2}\sum J_{ab} d\eta^a \wedge d\eta^b$$

−The form ω is closed: $\mathbf{d}\omega = \mathbf{dd}q^a \wedge \mathbf{d}p^a - \mathbf{d}q^a \wedge \mathbf{dd}p^a = 0$.

Poincaré theorem:

−If the phase space is single-connected, then ω is *exact* (exists 1-form h such that $\omega = \mathbf{d}h$)

Lemma:

−The form ω satisfies $\mathcal{L}_U\omega = 0$, where vector field

$$\boldsymbol{U} = \dot{\mathfrak{q}}_a(t)\frac{\partial}{\partial q^a} + \dot{\mathfrak{p}}_a\frac{\partial}{\partial p^a}$$

$\{\mathfrak{q}_a(t), \mathfrak{p}_a(t)\}$ is the trajectory, and we defined "Lie derivative" of *any* scalar function $f(t, q, p)$ from

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \dot{\mathfrak{q}}_a(t)\frac{\partial f}{\partial q^a} + \dot{\mathfrak{p}}_a\frac{\partial f}{\partial p^a} = \frac{\partial f}{\partial t} + \boldsymbol{U}f = \frac{\partial f}{\partial t} + \underbrace{\mathcal{L}_U f}_{\text{Lie derivative}}$$

Lemma (proof)

$$\mathcal{L}_U\omega = 0,$$

Proof:
►$\forall U$, ω we have Cartan's formula (also known as Cartan's theorem):

$$\mathcal{L}_{\boldsymbol{U}}\omega = [\mathbf{d}\omega(U) + \mathbf{d}\,(\omega \cdot U)]\,;$$

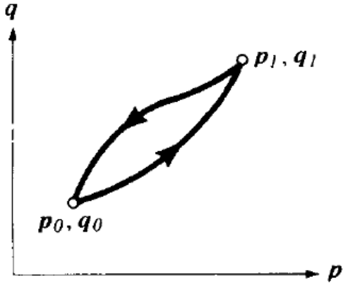
since $\mathbf{d}\omega = 0$, then

$$\begin{aligned}\mathcal{L}_{\boldsymbol{U}}\omega &= \mathbf{d}\,(\omega \cdot U) = \mathbf{d}\,(\dot{\mathfrak{q}}_a(t)dp^a - \dot{\mathfrak{p}}_a dq^a) \\ &= \mathbf{d}\left(\frac{\partial H}{\partial p^a}dp^a + \frac{\partial H}{\partial q^a}dq^a\right) = \mathbf{dd}H \equiv 0\end{aligned}$$

Q.E.D.
−Also conclude that $\omega \cdot U \equiv \mathbf{d}H$. We'll call any field U which has this property *Hamiltonian vector field.*
−Since the symplectic form $\omega = dq \wedge dp$ does not depend explicitly on time, $\partial_t\omega = 0$, this lemma implies that $d\omega/dt = 0$ along the particle's trajectory.

Example 4

Assume that we have a closed contour ∂S in the phase space formed by two trajectories in the phase space:



Demonstrate that the integral $\oint \boldsymbol{p} d\boldsymbol{q}$ remains invariant under time-independent canonical transformation

$$\oint_{\partial S} \boldsymbol{p} d\boldsymbol{q} = \oint_{\partial S^*} \boldsymbol{P} d\boldsymbol{Q}$$

Example 4 (proof)

Note that $\boldsymbol{d}\left(\sum_a p^a dq^a\right) = -\sum_a dq_a \wedge dp_a = -\omega$ (symplectic form), and also due to Stokes' theorem

$$\oint_{\partial S} \boldsymbol{p} d\boldsymbol{q} = \int_S \boldsymbol{d}\left(\sum_a p^a dq^a\right) = -\int_S \omega$$

Since ω is invariant under canonical transformations, $\omega = \sum_a dq_a \wedge dp_a = \sum_a dQ_a \wedge dP_a$, we may apply again Stoke's theorem and get

$$\oint_{\partial S} \boldsymbol{p} d\boldsymbol{q} = -\sum_a \int_{S^*} dQ_a \wedge dP_a = \oint_{\partial S^*} \boldsymbol{P} d\boldsymbol{Q}$$

$$\mathbf{d}\left(\boldsymbol{p} d\boldsymbol{q} - \boldsymbol{P} d\boldsymbol{Q}\right) = 0$$

so using Poincare's theorem conclude that it is exact,

$$\boldsymbol{p} d\boldsymbol{q} - \boldsymbol{P} d\boldsymbol{Q} = dF$$

where F is arbitrary function

−If we choose $F = F(\boldsymbol{q}, \boldsymbol{Q})$, get $p_a = \partial F / \partial q_a$ and $P_a = -\partial F / \partial Q_a$

−"Strange" receipt which allows to relate the coordinates (p, q) and (P, Q) via some unknown function (we'll see its meaning in the next slide)

Generating function

►We saw how to *check* rapidly that transformation is canonical

►Now we'll see how to *construct* the canonical transformations which automatically satisfies all the requirements

- Minimal action principle should be valid in any set of coordinates, (p, q) or (P, Q)

$$\delta S = \delta \int dt \left(\sum_a p_a \dot{q}_a - H(p,\, q,\, t) \right) =$$

$$= \delta \int dt \left(\sum_a P_a \dot{Q}_a - \tilde{H}(P,\, Q,\, t) \right) = \min$$

However in general we can't assume that integrands are equal

−they might differ by factor dF/dt

*as You remember adding total derivative to lagrangian doe not affect equations of motion

$$\Rightarrow \left(\sum_a p_a \dot{q}_a - H(p,\, q,\, t) \right) =$$

$$\left(\sum_a P_a \dot{Q}_a - \tilde{H}(P,\, Q,\, t) \right) + \frac{d}{dt} F(q,\, Q,\, t)$$

Generating function

$$\Rightarrow \left(\sum_a p_a \dot{q}_a - H(p,\, q,\, t)\right) = \left(\sum_a P_a \dot{Q}_a - \tilde{H}(P,\, Q,\, t)\right) + \frac{d}{dt} F(q,\, Q,\, t)$$

where $F(q,\, Q,\, t)$ is some function
$-$May rewrite this as:

$$dF = \sum p_a dq_a - \sum P_a dQ_a + \left(\tilde{H} - H\right)\, dt$$

$-$Conclude that

$$\Rightarrow p_a = \frac{\partial F}{\partial q_a}, \quad P_a = -\frac{\partial F}{\partial Q_a}, \quad \tilde{H} = H + \frac{\partial F}{\partial t}$$

$-$Function $F(q,\, Q)$ is called **generating function**, it allows to relate (implicitly) old and new coordinates

Example

Assume that the generating function $F_1(q, Q, t) = \sum_i q_i Q_i$. Write out explicitly the canonical transformation

Solution:

$$p_i = \frac{\partial F_1}{\partial q_i} = Q_i, \; P_i = -\frac{\partial F_1}{\partial Q_i} = -q_i$$

so can rewrite this as

$$Q_i = p_i, \; P_i = -q_i$$

Generating function

$$dF = \sum p_a dq_a - \sum P_a dQ_a + \tilde{H}\,dt - H\,dt$$

- Note that can rewrite

$$p_a dq_a = d\left(p_a q_a\right) - q_a dp_a$$

Let's introduce now

$$d\tilde{F} = \sum q_a dp_a - \sum P_a dQ_a + \tilde{H}\,dt - H\,dt$$

$$\tilde{F} = \tilde{F}\left(p,\ Q,\ t\right) = F - \sum_a p_a q_a$$

$\Rightarrow$We could use $\tilde{F}$ as another generating function which depends on variables p, Q, t

$$P_a = -\frac{\partial \tilde{F}}{\partial Q_a}, \quad q_a = \frac{\partial \tilde{F}}{\partial p_a}, \quad \tilde{H} = H + \frac{\partial \tilde{F}}{\partial t}$$

Example

Write out the canonical transformations (1) for the case when generating functional F depends on other possible combinations of old and new variables: $F_2\,(q,\ P,\ t)$, $F_3\,(p,\ Q,\ t)$, $F_4\,(p,\ P,\ t)$.

$$\Rightarrow p_a = \frac{\partial F_2}{\partial q_a},\ \ Q_a = \frac{\partial F_2}{\partial P_a} \qquad (2)$$

$$\Rightarrow q_a = -\frac{\partial F_3}{\partial p_a},\ \ P_a = -\frac{\partial F_3}{\partial Q_a} \qquad (3)$$

$$\Rightarrow q_a = -\frac{\partial F_4}{\partial p_a},\ \ Q_a = \frac{\partial F_4}{\partial P_a} \qquad (4)$$

Example - 1

Find the generating function of the *identical* canonical transformation

$$q_a = Q_a, \quad p_a = P_a$$

Assume that corresponding generating function is $F_2(q, P, t)$
►Correponding canonical transformations:

$$\Rightarrow p_a = \frac{\partial F_2}{\partial q_a} = P_a, \; Q_a = \frac{\partial F_2}{\partial P_a} = q_a$$

Check that equations are compatible:

$$\frac{\partial^2 F_2}{\partial q_a \partial P_b} = \frac{\partial^2 F_2}{\partial P_b \partial q_a} = \delta_{ab}$$

⇒transformation is canonical (as expected), so can solve them together:

$$\Rightarrow F_2 = q_a P_a + \mathrm{const}$$

Example 2

Find the values of α, β for which the transformation

$$q_a = \alpha Q_a, \quad p_a = \beta P_a$$

is canonical and find the generating function for this case

►Correponding canonical transformations:

$$\Rightarrow p_a = \frac{\partial F_2}{\partial q_a} = \beta P_a, \;\; Q_a = \frac{\partial F_2}{\partial P_a} = \frac{1}{\alpha} q_a$$

Request that equations are compatible:

$$\frac{\partial^2 F_2}{\partial q_a \partial P_a} = \frac{\partial^2 F_2}{\partial P_a \partial q_a} \Rightarrow \beta = \frac{1}{\alpha}$$

⇒can solve them together for $\beta = 1/\alpha$:

$$\Rightarrow F_2 = \frac{1}{\alpha} q_a P_a + \mathrm{const}$$

Generating function

Find the generating function of the canonical transformation

$$Q = \log\left(\frac{\sin p}{q}\right), \quad P = q \cot p$$

Solution:
We already checked that transformation is Canonical, so focus on construction of generating function
−From definition of canonical transformation:

$$p = \arcsin\left(q\, e^{Q}\right) = \frac{\partial F\,(q,\ Q,\ t)}{\partial q}, \quad P = \sqrt{e^{-2Q} - q^2} = -\frac{\partial F\,(q,\ Q,\ t)}{\partial Q}$$

Can check that these equations are compatible, evaluation of the second derivative $\partial^2 F/\partial q \partial Q$ from both gives the same result:

$$\frac{\partial^2 F\,(q,\ Q,\ t)}{\partial q \partial Q} = \frac{q\, e^{Q}}{\sqrt{1 - q^2 e^{2Q}}} \Rightarrow \text{system is compatible}$$

$$F = \int dq\, \frac{\partial F\,(q,\ Q)}{\partial q} = q \arcsin\left(q\, e^{Q}\right) + \sqrt{e^{-2Q} - q^2}$$

Example: Harmonic oscillator

Assume that the generating function is given by

$$F(q,\, Q) = \frac{m\omega q^2}{2} \cot Q$$

Find explicitly the canonic transformation which it generates and the hamiltonian of harmonic oscillator $H(P,\, Q)$ after this transformation

Caution: We'll see that transformation is NOT trivial and allows to drastically simplify analysis/solution of equations of motion

Example: Harmonic oscillator

$$\text{Hamiltonian}: \quad H = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}$$

$$\text{Transformation}: \quad p = \frac{\partial F}{\partial q} = m\omega q \cot Q, \quad P = -\frac{\partial F}{\partial Q} = \frac{m\omega q^2}{2\sin^2 Q}$$

$$\text{"solve" algebraically to get } (q,p): \quad q = \sqrt{\frac{2P}{m\omega}} \sin Q, \quad p = \sqrt{2m\omega P} \cos Q$$

Hamiltonian after transformation:

$$H_{\text{new}}(P,Q) = H_{\text{old}}\left(p(P,Q),\, q(P,Q)\right) + \underbrace{\frac{\partial F}{\partial t}}_{=0} = \omega P$$

-In new variables there is no dependence on Q at all, *i.e.* Q is cyclic variable !!!

$$\text{equations of motion}: \dot{P} = -\frac{\partial H_{\text{new}}}{\partial Q} = 0, \quad P = \text{const} = E/\omega$$

$$\dot{Q} = \frac{\partial H_{\text{new}}}{\partial P} = \omega \quad \Rightarrow Q = \omega t + \text{const}$$

Example: Harmonic oscillator

$$\text{Transformation}: \quad q = \sqrt{\frac{2P}{m\omega}}\sin Q, \quad p = \sqrt{2m\omega P}\cos Q$$

$$\text{Trajectory}: \quad P = \text{const} = E/\omega, \quad Q = \omega t + \text{const} \qquad (1)$$

Compare with result (p, q) variables:

$$q = \sqrt{\frac{2E}{m\omega^2}}\sin(\omega t + \phi_0), \quad p = \sqrt{2mE}\cos(\omega t + \phi_0) \Rightarrow \text{agrees with } (1)$$

–Use of canonical transformation drastically simplified solution: instead of coupled ODEs

$$\dot{p} = -m\omega^2 q, \quad \dot{q} = p/m \qquad \left[\ddot{q} + \omega^2 q = 0\right]$$

solved MUCH simpler decoupled system (1a,1b).

*Such variables (P, Q) are called "action-angle" variables; please see Goldstein, Landau for more details.

**Exist for any system (even multidimensional)

**Drastically simplify analysis: always Q_a are cyclic:

$$H = H(P_1, \ldots, P_N)$$

$$\dot{Q}_a = \text{const} = \alpha_a \quad \Rightarrow \quad Q_a = \alpha_a t + \text{const}$$

Action-angle variables (formal definitions)

Consider conservative system moving in potential field $U(q)$. Assume that the trajectory is closed, and define:

$$S_0\left(q,\, E\right) = \int^{q} p\, dq, \quad I\left(E\right) = \oint \frac{p\, dq}{2\pi}$$

$-S_0$ is the abbreviated action defined earlier
$-p$ is momentum. For conservative system $E = \mathrm{const}$, so

$$p = p(q, E) = \sqrt{2m(E - U(q))}$$

$-$Let's consider now $S_0\left(q,\, E\right) = S_0\left(q,\, I\right)$ as a generating function of a canonical transformation $F_2(q, P)$, in which I plays the role of new canonical momentum P:

$$p = \frac{\partial F_2}{\partial q} = \frac{\partial S_0}{\partial q}, \quad Q \equiv w = \frac{\partial F_2}{\partial I} = \frac{\partial S_0}{\partial I}$$

$-$variable I is called <u>*action*</u>, w is <u>*angle*</u>

Action-angle variables (formal definitions)

Consider conservative system, assume that the trajectory is closed, and define:

$$S_0\left(q,\, E\right) = \int^{q} p\, dq, \quad I\left(E\right) = \oint \frac{p\, dq}{2\pi}, \quad Q \equiv w = \frac{\partial S_0}{\partial I}$$

- variable I is called <u>*action*</u>, w is <u>*angle*</u>
- Since Hamiltonian (energy E) depends only on canonical momentum but not on coordinate w, the variable w is cyclical, and

$$\dot{I} = -\frac{\partial H}{\partial w} = 0, \quad \dot{w} = \frac{dH}{dI} = \frac{dE}{dI} = \mathrm{const}$$

$$\Rightarrow w = \frac{dE}{dI} t + \mathrm{const}$$

We'll show now that the canonical transformation which we applied earlier to Harmonic oscillator is just transformation to action-angle variables.

Harmonic oscillator in action-angle variables

$$\text{abbreviated action(gener. function)}: \; S_0 = \int \sqrt{2m(\omega P - kq^2/2)}dq$$

$$p = \frac{\partial S_0}{\partial q} = \sqrt{2m\left(\omega P - \frac{k\,q^2}{2}\right)}, \qquad Q = \frac{\partial S_0}{\partial P} = \omega\frac{\partial S_0}{\partial E} = \arctan\left(\frac{\sqrt{k\,q^2}}{\sqrt{2\omega P - k\,q^2}}\right),$$

$$\Rightarrow \tan Q = \frac{\sqrt{k\,q^2}}{\sqrt{2\omega P - k\,q^2}} = \frac{q}{p}\sqrt{2km}, \quad q = \frac{p}{\sqrt{km}}\tan Q = \frac{p}{m\omega}\tan Q$$

$$\frac{p^2}{2m} = \left(\omega P - \frac{k\,q^2}{2}\right) = \left(\omega P - \frac{p^2}{2m}\tan^2 Q\right)$$

$$\frac{p^2}{2m}\left(1 + \tan^2 Q\right) = \frac{p^2}{2m\cos^2 Q} = \omega P$$

$$\Rightarrow q = \frac{p}{m\omega}\tan Q = \sqrt{\frac{2P}{m\omega}}\sin Q, \qquad p = \sqrt{2m\omega P}\cos Q,$$

in agreement with our previous findings

Summary

Today we:

- **Canonical transformation:** any transformation which:
  - doesn't change canonical equation

$$\dot{p}_a = -\frac{\partial H(p,\, q, t)}{\partial q_a} = [H,\, p_a]\,, \qquad \dot{q}_a = \frac{\partial H(p,\, q, t)}{\partial p_a} = [H,\, q_a]\,.$$

  - doesn't change structure of J_{ab} and symplectic form

$$\omega = \frac{1}{2}\sum_{ab} J_{ab} d\eta_a \wedge d\eta_b = \sum_a dq_a \wedge dp_a$$

- Possible to generate the canonical transf. using generating functions, e.g.

$$\Rightarrow p_a = \frac{\partial F}{\partial q_a}, \quad P_a = -\frac{\partial F}{\partial Q_a}, \quad \tilde{H} = H + \frac{\partial F}{\partial t}$$

- Canonical transformations can drastically simplify hamiltonian and dynamics (*action-angle variables for harmonic oscillator*)

In the next lectures we'll see more examples where canonical transformations simplify analysis